MIVA
OPEN UNIVERSITY

Course Name

Computer Programming I

Topic: Logical expressions and Boolean operators

Week 3

BULLET POINT SUMMARY

This week, some of the learning points include:

- Logic forms the foundation for decision-making and programme flow, making it essential for intelligent software.
- Logical expressions evaluate to true or false, enabling programmes to make choices based on conditions.
- The boolean data type represents true or false values, named after George Boole and crucial for logical expressions.
- Basic Boolean Operators:
- - AND (&&): true only if both operands are true. (e.g., user has valid username and password.)
- - OR (||): true if at least one operand is true. (e.g., user has membership or purchase exceeds amount)
- - NOT (!): Negates a value (e.g., user not logged in)
- Complex logical expressions combine multiple operators for intricate conditions, and parentheses define evaluation order.
- The order of operations of logical operators is arithmetic > comparison > logical, with left-to-right evaluation and short-circuit optimisation.
- Real-world examples include login systems, navigation apps, age restrictions, and online discounts.
- Logical expressions and Boolean operators are powerful tools for creating dynamic, intelligent, and decision-making programmes.



PERSONAL REFLECTION

Reflect on a recent programming challenge where you used logical expressions and boolean operators.

- Did you encounter any difficulties in understanding or implementing them?
- How did you approach debugging your code and ensuring its logical correctness?

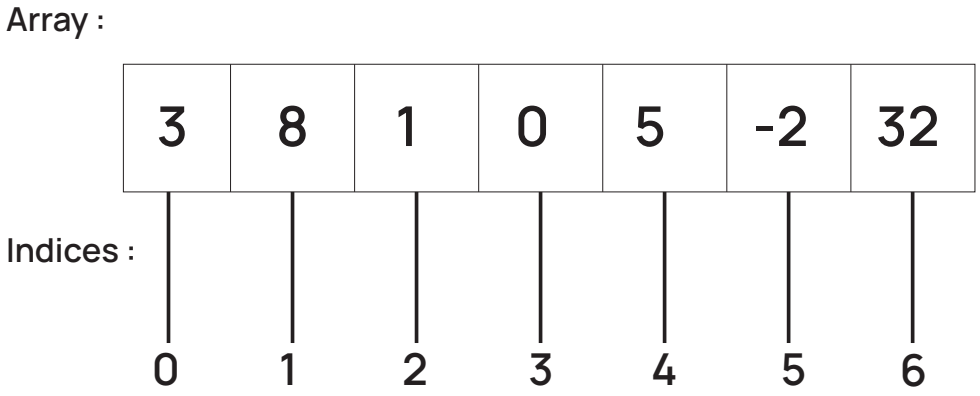
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Key references:
Sierra, K., & Bates, B. (2005). Head First Java: A Brain-Friendly Guide. " O'Reilly Media, Inc.".
Horstmann, C. S. (2000). Core JAVA Vol Fundamentals.

CALL TO ACTION

Miva University wants to upgrade to smart home technology. As a computing student, you are tasked with designing a smart home system that adjusts lighting and temperature based on user preferences and real-time data. Also take into consideration planning different logical conditions, prioritising user comfort and efficiency, and considering edge cases like sensor malfunctions or sudden weather changes.

- How would you approach designing the logical expressions to control lights and temperature efficiently?
- What factors would you consider in terms of user comfort, energy consumption, and edge case handling?



SKILLS AND COMPETENCIES YOU HAVE ACQUIRED AFTER THIS LESSON

- Logical reasoning
- Enhanced analytical skills
- Enhance coding accuracy



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Key references:

Burd, B. (2017). Beginning programming with Java for Dummies. John Wiley & Sons.
Eckel, B. (2003). Thinking in JAVA. Prentice Hall Professional.